

# PRITOM MAJUMDER

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## RESEARCH INTEREST

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- **Functional, Piezoelectric Ceramics & Nanomaterials:** Lead-free ferroelectrics, solid-state synthesis, dielectric and piezoelectric response, magnetic ceramics and low-temperature sintering routes.
- **Oxide Semiconductor Thin Films:** Doping-driven control of structural, optical, magnetic and electronic behaviour in transparent conducting oxides.
- **Photonic & Optoelectronic Materials:** Advanced ceramics, lasers, LEDs, photovoltaics, photonics and magnetic data storage and optical fibers.
- **Metallurgy:** Corrosion and surface engineering, ferrous metallurgy, materials characterization and mechanical behaviour of materials.
- **Computational Material Science:** Density Functional Theory (DFT)

## EDUCATION

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### B.Sc. in Materials and Metallurgical Engineering

2021 to June 2026

Bangladesh University of Engineering and Technology (BUET), Dhaka

CGPA: 3.64 (15<sup>th</sup> of 59) / 4.00

**Relevant coursework:** Advanced Ceramics (A+); Nanostructured Materials & Thin Films (A+); Electrical, Optical and Magnetic Properties of Materials (A+); Characterisation of Materials Laboratory (A+); Corrosion and Degradation of Metals and Alloys (A+); Surface Engineering and Tribology (A+); Photonic and Magnetic Materials (A).

## SELECTED PROJECTS

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### [Cu-Doped SnO<sub>2</sub> Thin Films](#) · [Solid State Physics Laboratory](#), BUET

2024 to 2026

Hydrothermal growth and defect engineering ; Supervisor: Dr. Md. Azizar Rahman ; Manuscript in preparation (first author)

- Targeted p-type Cu:SnO<sub>2</sub> for an all-SnO<sub>2</sub> p-n homojunction diode, a route that collaborator-run DFT+U calculations indicated would require the oxygen vacancies to be suppressed.
- Grew undoped and Cu-doped (5, 10 and 15 at%) SnO<sub>2</sub> thin films directly on soda-lime glass by a single-step hydrothermal route at 180 °C, using NaOH and NaCl as growth mediators to tailor the defect chemistry during nucleation.
- Stabilized the metastable orthorhombic ( $\alpha$ -PbO<sub>2</sub>-type) polymorph at ambient pressure, confirmed by XRD, HRTEM, SAED and EDS. Tracked the doping-driven band gap by UV-Vis.
- Established from I-V and Mott-Schottky analysis of the Cu:SnO<sub>2</sub>/ITO junction that Cu doping does not invert the conductivity type (remained n-type).

### [Millsorpt - Capstone Design Project](#) · Department of MME, BUET

2025

- Converted ferrous mill scale, a low-value steel-industry waste, into a chemically treated adsorbent (hematite rich iron-oxide) for heavy metal wastewater treatment
- Designed a low-cost bench scale filtration column achieving a high removal of Cu, Cr and Ni from mixed-metal solutions; iron-oxide phases confirmed by XRD, metal removal quantified by AAS and UV-Vis.
- Presented the work to specialists at Abul Khair Steel, BCSIR, the Bangladesh Atomic Energy Commission etc., receiving positive technical feedback.

### [3D PotterBot - CAD Modelling](#) · SolidWorks

2023

- Modelled a paste-extrusion ceramic 3D printer as a fully-mated 118 parts assembly; produced photorealistic renders, an exploded-view motion study, and a static Finite Element Analysis of the vertical support rod.

## RESEARCH EXPERIENCE

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### Undergraduate Thesis Researcher · [Ahmed Sharif's Lab](#) · Department of MME, BUET

2025 to 2026

"Investigation of Dielectric and Piezoelectric Properties of KNN with BCHfT Composite Ceramics" ; Supervisor: Prof. Ahmed Sharif ; [THESIS BOOK & THESIS PPTX](#)

- Synthesized lead-free K<sub>0.5</sub>Na<sub>0.5</sub>NbO<sub>3</sub> (KNN) ceramics modified with Ba<sub>0.85</sub>Ca<sub>0.15</sub>Hf<sub>0.1</sub>Ti<sub>0.9</sub>O<sub>3</sub> (BCHfT) by the conventional solid-state route.

- Characterized phase evolution and microstructure by XRD and FE-SEM; measured frequency- and temperature-dependent dielectric permittivity and loss, impedance spectroscopy, and the piezoelectric charge coefficient  $d_{33}$ .
- Identified KNN-1wt% BChfT as the optimum composition, giving the best balance of dielectric response.

**Undergraduate Researcher** · [Solid State Physics Laboratory](#) · Department of Physics, BUET **2024 to 2026**

- Synthesis of SnO<sub>2</sub> and Cu:SnO<sub>2</sub> research by hydrothermal process.

**Undergraduate Thesis Researcher** · [Pilot plant and Process Development Center](#) · BCSIR **2025 to 2026**

- Carried out the complete characterization of the KNN-BChfT thesis series on BCSIR instruments: XRD, SEM, frequency- and temperature-dependent dielectric test.

## ON GOING WORK

1. **Pritom Majumder**, Anabil Dey, and Ahmed Sharif. "Advancements in Cold Sintering of Piezoceramics: Flux Mediation, Microstructural Evolution, and Enhanced Dielectric Properties - A Review." (under peer review) Journal: Emergent Materials.

## INDUSTRIAL AND LABORATORY TRAINING

[Bangladesh Council of Scientific and Industrial Research \(BCSIR\)](#) · Dhaka **2026**

[Abul Khair Steel Melting Limited \(AKSML\)](#) · Chittagong **March 2025**

[Bangladesh Industrial Technical Assistance Centre \(BITAC\)](#) · Dhaka **March 2025**

## TECHNICAL SKILLS

- **Synthesis, Processing and Data Analysis:** Solid-state ceramic synthesis; hydrothermal synthesis; sintering; uniaxial pressing; XRD; FE-SEM and SEM; TEM including HRTEM, SAED; Impedance spectroscopy; frequency- and temperature-dependent dielectric measurement;  $d_{33}$  meter; cyclic voltammetry; I-V and Mott-Schottky analysis; UV-Vis spectroscopy; photoluminescence (PL); FTIR
- **Mechanical & Chemical Testing:** Tensile, Charpy impact and three-point flexural testing; Vickers, Rockwell and Brinell hardness; optical emission spectroscopy (OES); gravimetric and wet chemical analysis
- **Software:** OriginPro; MATLAB; SolidWorks; ImageJ; X'Pert HighScore; C/C++; MS Office; Adobe Photoshop and Illustrator

## AWARDS AND SCHOLARSHIPS

- **University Merit Scholarship**, Bangladesh University of Engineering and Technology **(July 2022)**
- **University Stipend Scholarship**, Bangladesh University of Engineering and Technology (July 2021 and January 2022)
- **Letter of Appreciation**, Abul Khair Steel Melting Limited - for professionalism and work ethic during industrial training **2025**
- **Certificate of Perfect Attendance**, Notre Dame College, Dhaka **(2019 to 2020)**

## LANGUAGE PROFICIENCY

**English:** IELTS Academic - Overall Band 7.0 (Listening 7.5, Reading 7.5, Writing 6.0, Speaking 6.5) **2026**

## LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- **Director of Administration**, Executive Committee, BUET Debating Club (BUETDC), 2025-26 :- previously Information Secretary. Organizing panel member for BUETDC Nationals 2024; finalist at the BUET Freshers' Debating Tournament; competed at the Inter-Faculty Tournament 2024 and the 6th GSCDC Nationals 2024.
- **Organizer**, Winter Fitness Competition 2025, BUET Gymnasium :- planned and ran the event in collaboration with the resident trainer; Organized Treasure-Hunt competition 2024, as an organizing panel member of "Reception'21"
- **Class Representative**, Department of MME, BUET, 2021-22 :- elected by classmates for both semesters of the first year
- **Active Member & Blood Donor**, Badhon - a non-profit voluntary blood donation organization, 2022-present